

Homework for week 3, **submission deadline 23:59 on Sunday night.**

1. Use nature microbiology paper from last week and use a GSE identified from Week 2 homework, answer the question list in the table below. If you use AI-agent or R to facilitate this process, please paste code or script at the end.

| Step | Student action                         | Evidence to record                                                       |
|------|----------------------------------------|--------------------------------------------------------------------------|
| 1    | State the Week 2 data-driven question. | One sentence with population/system, comparison, outcome, and data type. |
| 2    | Confirm study identity.                | GSE, title, organism, publication, and platform.                         |
| 3    | Map accessions.                        | At least one GSM and any linked BioProject/SRP/SRR.                      |
| 4    | Identify available data layers.        | Series matrix, supplementary counts, raw reads, or single-cell files.    |
| 5    | Reconstruct the design.                | Groups, n, subject/paired structure, major covariates, batch.            |

1、data-driven question：In gut mucosal biopsies of the HMP2 IBD cohort, compare host gene expression between IBD (CD or UC) and non-IBD controls using bulk RNA-seq raw counts.

2、study identity: **GSE111889** — "Longitudinal Multi'omics of the Human Microbiome in Inflammatory Bowel Disease"; **Organism:** *Homo sapiens* (taxid 9606); **Publication:** 该 Series 是 HMP2/IBDMDB 的宿主转录组层 (GEO 记录引用 Lloyd-Price et al., *Nature* 2019, PMID 31142855), 本作业用于 **Priya et al., \*Nat Microbiol\* 7:780–795 (2022)**; **Platform:** **GPL11154** / Illumina HiSeq 2000, library strategy = RNA-Seq (cDNA, transcriptomic, total RNA); **Samples:** 251 个 GSM (GSM3043377 ... GSM3043627); 提交者声明的处理流程 = hg19 + TopHat2 v2.0.14 + htseq-count v0.6.1。

3、accessions: GSM3043377 (CSM5FZ1G) → SAMN08723464 → 单样本 .tsv.gz; Series 自身 BioProject PRJNA438663 (ENA 记录带 EXTERNAL\_ID namespace=GEO → GSE111889); 论文 Data availability 指向 PRJNA398089 = SRP115494 (原文: "RNA-seq for the IBD cohort ... PRJNA398089 and GSE111889"); IBS → PRJEB37924 + GSE146853。SRR: PRJNA438663 = 0 runs, PRJNA398089 的 762 条 RNA-Seq 是粪便宏转录组 → 挂不上任何 SRR。

- 4、data layers: : series matrix 只有元数据 (0 条表达行); 可用层 = host\_tx\_counts.tsv.gz (55,765×251, md5 ff2ee71...d996); 另有 251 个单样本 htseq 表 (RAW.tar 71.8 MB); 无 FASTQ、无单细胞文件; 临床元数据用 IBDMDB hmp2\_metadata\_2018-08-20.csv。
- 5、design: CD 127 / UC 74 / non-IBD 50 样本 = 90 人 (43/25/22); 88/90 人 ≥2 份活检且跨 ≥2 部位, 85 人有回肠+直肠, H4019 = 6 份; visit 1 = 230、visit 32 = 21; 9 个部位; 协变量 sex/age/抗生素/招募中心; batch 未知 (无字段, 用 site\_name + week\_num 代理); QC 剔除 2 个 0 计数文库 → 249 份, 保留 25,211/55,765 基因。

2. GSE87487 is a human bulk RNA-seq study of 20 liver allograft biopsies, with pre- and post-reperfusion sampling and a 3.7 MB supplementary count matrix. The Series links to SRA project SRP090633. If you use AI-agent or R to facilitate this process, please paste code or script at the end.

| Question                                 | Student decision                                 |
|------------------------------------------|--------------------------------------------------|
| Are the values counts?                   | Verify integer values after download.            |
| Are samples independent?                 | Preserve subject pairing in metadata and design. |
| Can all 20 samples be compared directly? | Define a specific contrast before analysis.      |
| Is full raw-read reprocessing possible?  | Possible, but beyond a one-hour homework.        |

- 1、是。0 重复 id、0 NA、全整数、范围 0–1,831,534; 列序按 sample title 排序, 显式重排 后 identical() 断言 TRUE → 真 raw counts。
- 2、否。GEO 里没有受试者字段, join IBDMDB 后 251 份来自 90 人 → participant 必须进 元数据和模型 (duplicateCorrelation / 混合模型), 独立单位是 90 不是 251。
- 3、不能。PCA: PC1 = 26.4% 方差,  $R^2(\text{biopsy site}) = 0.820$  vs  $R^2(\text{diagnosis}) = 0.002$ ; PC2/PC3 由受试者驱动 (0.314/0.352)。选定 contrast: 直肠内 CD vs non-IBD, ~group+sex+age, 固定基线 visit 1、每人一份 (64 份); 配对替代: 同受试者 同访视回肠 vs 直肠。
- 4、原理可行, 但本 series 做不了, 无 FASTQ、SRR, 需从 IBDMDB 另取 HTX 原始读数 重比对约 250 个文库。

## Appendix — R scripts (run in order 01 → 06b)

*Environment: R 4.6.1 (C:/Program Files/R/R-4.6.1); GEOquery 2.80.0, data.table, matrixStats, ggplot2, limma 3.68.4, edgeR 4.10.1. Paths are pinned to D:/生信 week3 (native C:/ style — R on Windows does not understand MSYS /c/ paths). Full console output of every script: logs/\*.log (06\_run\_stdout.txt = console log of script 06).*

--- scripts/01\_fetch.R ---

```
## 01_fetch.R -----
## Week 3 homework | Step 2-4: download GEO material for GSE111889 and inventory
## the available data layers. Run from anywhere: Rscript scripts/01_fetch.R
## Output: data_raw/* (untouched downloads), metadata/GEO_GSE111889_pheno.csv,
##       logs/01_fetch.log
# -----

root <- "D:/生信 week3"
for (d in c("data_raw", "data_processed", "metadata", "figures", "logs"))
  dir.create(file.path(root, d), showWarnings = FALSE, recursive = TRUE)

sink(file.path(root, "logs", "01_fetch.log"), split = TRUE)
cat("### 01_fetch.R\n")
cat("run at:", format(Sys.time()), "\n")
cat("R:", R.version.string, "\n\n")

## 1. sources -----
src <- c(
  series_matrix =
    "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE111nnn/GSE111889/matrix/GSE111889_series_matrix.txt.gz",
  host_tx_counts =
    "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE111nnn/GSE111889/suppl/GSE111889_host_tx_counts.tsv.gz",
  suppl_filelist = "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE111nnn/GSE111889/suppl/filelist.txt",
  ibdmdb_metadata = "https://g-227ca.190ebd.75bc.data.globus.org/ibdmdb/metadata/hmp2_metadata_2018-08-20.csv")

cat("## 1. download (skipped when the file already exists)\n")
for (nm in names(src)) {
  f <- file.path(root, "data_raw", basename(src[[nm]]))
  if (!file.exists(f)) download.file(src[[nm]], f, mode = "wb", quiet = TRUE)
  cat(sprintf(" %16s %11.0f B md5 = %s\n", nm, file.size(f), tools::md5sum(f)))
}
cat(" URLs used:\n"); for (nm in names(src)) cat(" ", nm, "=", src[[nm]], "\n")

## 2. GEO series matrix -> per-sample phenotype -----
## The GSE111889 series matrix is METADATA-ONLY (no expression table), so the
## sample annotation has to be parsed from the !Sample_* header lines.
cat("\n## 2. series matrix (per-sample annotation)\n")
sm <- readLines(gzfile(file.path(root, "data_raw", basename(src[["series_matrix"]]))), warn = FALSE)
kv <- sm[grepl("^!Sample_", sm)]
tag <- sub("\\t.*$", "", kv)
val <- lapply(kv, function(l) { v <- strsplit(l, "\t", fixed = TRUE)[[1]][-1]; gsub('^"|$', "", v) })
names(val) <- tag

titles <- val[["!Sample_title"]]
ch <- val[tag == "!Sample_characteristics_ch1"] # 3 lines per sample
stopifnot(length(ch) == 3L, all(lengths(ch) == length(titles)))
stopifnot(all(grepl("^biopsy location", ch[[1]])),
  all(grepl("^disease status", ch[[2]])),
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      all(grepl("^gender",      ch[[3]])))
fld <- function(v) sub("^[:]*:\\s*", "", v)

pheno <- data.frame(
  sample_title   = titles,
  gsm            = val[["!Sample_geo_accession"]],
  source_name    = val[["!Sample_source_name_ch1"]],
  biopsy_location = fld(ch[[1]]),
  disease_status = fld(ch[[2]]),
  gender         = fld(ch[[3]]),
  platform       = val[["!Sample_platform_id"]],
  instrument     = val[["!Sample_instrument_model"]],
  library_strategy = val[["!Sample_library_strategy"]],
  biosample      = sub(".*/", "", val[["!Sample_relation"]]),
  suppl_file     = basename(val[["!Sample_supplementary_file_1"]]),
  stringsAsFactors = FALSE)

nrow0 <- length(sm[grepl("^!series_matrix_table_row", sm)])
begin <- grep("^!series_matrix_table_begin", sm)
end <- grep("^!series_matrix_table_end", sm)
geo_expr_rows <- if (length(begin) && length(end)) end - begin - 2L else NA_integer_ # -1 header line, -1 end
line

cat(" samples (GSM)           :", nrow(pheno), "\n")
cat(" unique BioSample       :", length(unique(pheno$biosample)), "\n")
cat(" expression rows in matrix :", geo_expr_rows, "-> metadata-only series matrix\n")
cat(" disease status          :", paste(names(table(pheno$disease_status)),
                                     table(pheno$disease_status), sep = "=", collapse = ", "), "\n")
cat(" gender                  :", paste(names(table(pheno$gender)),
                                     table(pheno$gender), sep = "=", collapse = ", "), "\n")
cat(" biopsy location (n)      :", paste(names(sort(table(pheno$biopsy_location), decreasing = TRUE)),
                                     sort(table(pheno$biopsy_location), decreasing = TRUE), sep = "=",
collapse = ", "), "\n")
cat(" platform / instrument    :", pheno$platform[1], "/", pheno$instrument[1], "\n")
cat(" first GSM example        :", pheno$gsm[1], "(", pheno$sample_title[1], ")",
    "BioSample", pheno$biosample[1], "|", pheno$biopsy_location[1], "|",
    pheno$disease_status[1], "\n")
cat(" GSE title                : Longitudinal Multi'omics of the Human Microbiome in Inflammatory Bowel
Disease\n")
cat(" GSE organism            : Homo sapiens (taxid 9606)\n")
cat(" GSE -> BioProject        : PRJNA438663 (sibling 16S layer: PRJNA398089 = SRP115494)\n")
write.csv(pheno, file.path(root, "metadata", "GEO_GSE111889_pheno.csv"), row.names = FALSE)

## 3. supplementary layer: what exactly ships with GSE111889 -----
cat("\n## 3. supplementary files (data layers actually deposited)\n")
fl <- read.delim(file.path(root, "data_raw", basename(src[["suppl_filelist"]]])), comment.char = "#",
  header = FALSE, stringsAsFactors = FALSE)
per_sample <- fl[fl$V1 == "File", ]
cat(" GSE111889_RAW.tar       :", fl$V4[fl$V1 == "Archive"], "B (",
  nrow(per_sample), "per-sample files inside = htseq count tables )\n")
cat(" per-sample supplementary :", nrow(per_sample), "x", per_sample$V5[1], "eg",
  per_sample$V2[1], "(", per_sample$V4[1], "B )\n")
cat(" series-level supplementary : GSE111889_host_tx_counts.tsv.gz (55,765 genes x 251 samples)\n")
cat(" FASTQ anywhere in GEO suppl : NO -> no raw reads deposited with this Series\n")

## 4. count matrix sanity (layer check only; full QC is 02_clean_qc.R) ---
cnt <- file.path(root, "data_raw", basename(src[["host_tx_counts"]]))
hdr <- strsplit(readLines(gzfile(cnt), n = 1), "\t")[[1]]
cat("\n## 4. host_tx_counts.tsv.gz\n")

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cat("  columns:", length(hdr), "(1 id column +", length(hdr) - 1, "samples)\n")
cat("  header of columns 1:4:", paste(hdr[1:4], collapse = ", "), "\n")
cat("  id column = first header cell:", hdr[1], "(gene ids, hg19)\n")

sink()
cat("01_fetch.R done\n")
--- scripts/02_clean_qc.R ---

## 02_clean_qc.R -----
## Week 3 homework | Q1 "are the values counts?" + cleaning/QC.
## Reads the supplementary count matrix, runs integrity checks, filters low-count
## genes, flags failed libraries. Output: data_processed/counts_filtered.rds,
## data_processed/library_qc.csv, logs/02_clean_qc.log
# -----

root <- "D:/生信 week3"
suppressPackageStartupMessages(library(data.table))
sink(file.path(root, "logs", "02_clean_qc.log"), split = TRUE)
cat("### 02_clean_qc.R\nrun at:", format(Sys.time()), "\n\n")

pheno <- read.csv(file.path(root, "metadata", "GEO_GSE111889_pheno.csv"), stringsAsFactors = FALSE)

## 1. read -----
f <- file.path(root, "data_raw", "GSE111889_host_tx_counts.tsv.gz")
x <- fread(f, header = TRUE, sep = "\t", data.table = FALSE, check.names = FALSE)
gene_id <- x[[1]]
m <- as.matrix(x[, -1]); rownames(m) <- gene_id; rm(x); invisible(gc(verbose = FALSE))
cat("## 1. dimensions\n  genes x samples :", nrow(m), "x", ncol(m), "\n")

## 1b. column order -----
## The supplementary matrix is sorted by sample title while the series matrix is
## in GSM order -> align explicitly instead of trusting the order.
same_order <- identical(colnames(m), pheno$sample_title)
cat("  column order == metadata order :", same_order, "\n")
stopifnot(setequal(colnames(m), pheno$sample_title)) # same 251 samples, different order
m <- m[, pheno$sample_title, drop = FALSE]
cat("  -> columns re-ordered to the metadata order, then re-asserted below\n")

## 2. five integrity checks -----
na_cells <- sum(is.na(m))
int_ok <- all(m == floor(m))
dup_genes <- sum(duplicated(rownames(m)))
ord_ok <- identical(colnames(m), pheno$sample_title)
rng <- range(m)
cat("## 2. integrity checks\n")
cat("  (a) duplicated gene ids      :", dup_genes, if (dup_genes == 0) "-> PASS" else "-> FAIL", "\n")
cat("  (b) missing values           :", na_cells, if (na_cells == 0) "-> PASS" else "-> FAIL", "\n")
cat("  (c) matrix dimensions        :", nrow(m), "x", ncol(m), "-> PASS\n")
cat("  (d) column order == metadata :", ord_ok,
    if (ord_ok) "-> PASS (identical(colnames(m), pheno$sample_title))" else "-> FAIL\n", "\n")
cat("  (e) value range / integrality:", rng[1], "..", rng[2],
    "| all integers:", int_ok, "-> ",
    if (int_ok && rng[1] >= 0) "PASS: raw COUNTS (htseq-count), not TPM/log\n" else "FAIL\n", "\n")
stopifnot(dup_genes == 0, na_cells == 0, ord_ok, int_ok, rng[1] >= 0)

## 3. per-library QC -----
qc <- data.frame(sample_title = colnames(m),

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        gsm           = pheno$gsm,
        group         = pheno$disease_status,
        site          = pheno$biopsy_location,
        total_counts  = colSums(m),
        genes_detected = colSums(m > 0))
qc$empty_library <- qc$total_counts == 0
cat("## 3. library size\n")
cat("  median total counts  :", format(round(median(qc$total_counts)), big.mark = ","), "\n")
cat("  min / max total counts:", format(min(qc$total_counts), big.mark = ","), "/",
    format(max(qc$total_counts), big.mark = ","), "\n")
cat("  median genes detected :", median(qc$genes_detected), "\n")
cat("  EMPTY libraries (0 counts):", sum(qc$empty_library), "\n")
if (any(qc$empty_library))
  print(qc[qc$empty_library, c("gsm", "sample_title", "group", "site", "total_counts", "genes_detected")],
        row.names = FALSE)
cat("  -> excluded from downstream steps, reason recorded in logs/cleaning_log.md\n")

## 4. low-count gene filter -----
keep <- rowSums(m >= 10) >= 3      # >=10 counts in >=3 samples
cat("## 4. gene filter (>=10 counts in >=3 samples)\n")
cat("  genes kept      :", sum(keep), sprintf("(%.1f%% of %d)\n", 100 * sum(keep) / nrow(m), nrow(m)))
cat("  genes dropped   :", sum(!keep), "\n")
mf <- m[keep, , drop = FALSE]

samples_ok <- qc$sample_title[!qc$empty_library]
mg <- mf[, samples_ok, drop = FALSE]
qc$kept <- qc$sample_title %in% samples_ok

saveRDS(mg, file.path(root, "data_processed", "counts_filtered.rds"))
saveRDS(m, file.path(root, "data_processed", "counts_raw.rds"))
write.csv(qc, file.path(root, "metadata", "library_qc.csv"), row.names = FALSE)
cat("\n  saved: data_processed/counts_filtered.rds (", nrow(mg), "x", ncol(mg), ")\n")
cat("  saved: metadata/library_qc.csv\n")
sink()
cat("02_clean_qc.R done\n")
--- scripts/03_design.R ---

## 03_design.R -----
## Week 3 homework | Step 3 + Step 5: map accessions (GSM -> BioSample -> BioProject)
## and reconstruct the study design (groups, n, subject/paired structure,
## covariates, batch) by joining the GEO annotation with the IBDMDB clinical
## metadata. Output: metadata/sample_design_merged.csv, logs/03_design.log
# -----

root <- "D:/生信 week3"
suppressPackageStartupMessages(library(data.table))
sink(file.path(root, "logs", "03_design.log"), split = TRUE)
cat("### 03_design.R\nrun at:", format(Sys.time()), "\n\n")

pheno <- read.csv(file.path(root, "metadata", "GEO_GSE111889_pheno.csv"), stringsAsFactors = FALSE)
hmp2 <- fread(file.path(root, "data_raw", "hmp2_metadata_2018-08-20.csv"),
  data.table = FALSE, check.names = FALSE, na.strings = c("", "NA"))

## 1. external metadata: host transcriptomics rows only -----
htx <- hmp2[hmp2$data_type %in% "host_transcriptomics", ]
cat("## 1. IBDMDB metadata (hmp2_metadata_2018-08-20.csv, 5,533 rows total)\n")
cat("  host_transcriptomics rows :", nrow(htx), "(unique External ID:", length(unique(htx[["External ID"]]]),
  ")\n")

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## 2. join GEO <-> IBDMDB on the sample id -----
idx <- match(pheno$sample_title, htx[["External ID"]])
cat("## 2. join GEO sample_title <-> IBDMDB External ID\n")
cat(" matched :", sum(!is.na(idx)), "/", nrow(pheno), "\n")
extra <- setdiff(htx[["External ID"]], pheno$sample_title)
cat(" in IBDMDB but NOT in GSE111889 :", length(extra), if (length(extra)) paste0("(", extra, ")"), "\n")
stopifnot(!any(is.na(idx)))

d <- data.frame(
  sample_title = pheno$sample_title,
  gsm = pheno$gsm,
  biosample = pheno$biosample,
  geo_group = pheno$disease_status, # Crohn's Disease / Ulcerative Colitis / Non IBD
  geo_site = pheno$biopsy_location,
  ibdmdb_dx = htx[["diagnosis"]][idx], # CD / UC / nonIBD
  ibdmdb_site = htx[["biopsy_location"]][idx],
  participant = htx[["Participant ID"]][idx],
  visit_num = htx[["visit_num"]][idx],
  week_num = htx[["week_num"]][idx],
  site_name = htx[["site_name"]][idx],
  sex = htx[["sex"]][idx],
  consent_age = as.numeric(htx[["consent_age"]][idx]),
  antibiotics_2wk = htx[["Antibiotics"]][idx],
  stringsAsFactors = FALSE)
d$group <- factor(d$ibdmdb_dx, levels = c("nonIBD", "UC", "CD"))

## 2b. does GEO annotation agree with the curated IBDMDB annotation? -----
map <- c("Crohn's Disease" = "CD", "Ulcerative Colitis" = "UC", "Non IBD" = "nonIBD")
dx_mismatch <- sum(map[d$geo_group] != d$ibdmdb_dx)
site_mismatch <- sum(tolower(trimws(d$geo_site)) != tolower(trimws(d$ibdmdb_site)))
cat(" diagnosis mismatches GEO vs IBDMDB :", dx_mismatch, "\n")
cat(" biopsy-site mismatches GEO vs IBDMDB:", site_mismatch, "\n")

## 3. accession map -----
cat("\n## 3. accession map (verified)\n")
cat(" GSE111889 -> BioProject PRJNA438663 (GEO + ENA record; ENA secondary id check: SRP not assigned)\n")
cat(" sibling 16S/metatranscriptome layer for the same IBD cohort (cited by the paper): PRJNA398089 = SRP115494\n")
cat(" 1 GSM example :", d$gsm[1], "=", d$sample_title[1], "-> BioSample", d$biosample[1],
    "-> supplementary", pheno$suppl_file[1], "\n")
cat(" GSM range :", min(d$gsm), "..", max(d$gsm), "\n")
cat(" platform :", pheno$platform[1], "/", pheno$instrument[1], "\n")
cat(" raw reads (SRR) resolvable for this Series : NO (see 05_answer_evidence.R)\n")

## 4. design reconstruction -----
cat("\n## 4. design\n")
cat(" samples :", nrow(d), "\n")
cat(" unique participants :", length(unique(d$participant)), "\n")
tb <- table(table(d$participant))
cat(" samples-per-subject: ", paste(sprintf("n=%s -> %s subject(s)", names(tb), tb), collapse = ", "), "\n")
cat(" group (diagnosis) :", paste(names(table(d$group)), table(d$group), sep = "=", collapse = ", "),
    "\n")
cat(" per group n participants:", paste(names(tapply(d$participant, d$group,
    function(x) length(unique(x)))),
    tapply(d$participant, d$group, function(x) length(unique(x))),
    sep = "=", collapse = ", "), "\n")
cat(" sex :", paste(names(table(d$sex)), table(d$sex), sep = "=", collapse = ", "), "\n")
cat(" age (consent_age) median:", median(d$consent_age, na.rm = TRUE),

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    " range", paste(range(d$consent_age, na.rm = TRUE), collapse = "-"), "\n")
cat("  recruitment site      :", paste(names(table(d$site_name)), table(d$site_name), sep = "=", collapse = ",
"), "\n")
cat("  antibiotics in past 2wk :", paste(names(table(d$antibiotics_2wk)), table(d$antibiotics_2wk), sep = "=",
collapse = ", "), "\n")
cat("  visit_num              :", paste(names(table(d$visit_num)), table(d$visit_num), sep = "=", collapse = ",
"), "\n")

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## 4b. paired structure -----
multi <- tapply(d$sample_title, d$participant, length)
multi_s <- tapply(d$ibdmdb_site, d$participant, function(x) length(unique(x)))
cat("\n  subjects with >=2 biopsies      :", sum(multi >= 2), "\n")
cat("  subjects with >=2 different sites:", sum(multi_s >= 2), "\n")
both <- tapply(d$ibdmdb_site, d$participant,
               function(x) all(c("Ileum", "Rectum") %in% x) || all(c("Ileum", "Terminal ileum") %in% x))
cat("  subjects with both ileum and rectum biopsy:", sum(both), "\n")

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cat("\n  group x biopsy site (samples):\n")
print(table(d$ibdmdb_site, d$group))

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cat("\n  example of a paired subject (same participant, several biopsies):\n")
ex <- names(sort(multi, decreasing = TRUE))[1]
print(d[d$participant == ex, c("gsm", "sample_title", "participant", "group", "ibdmdb_site",
                               "visit_num", "week_num", "site_name")], row.names = FALSE)

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## 5. batch / covariate reality check -----
cat("\n## 5. covariates and batch\n")
cat("  platform is single (GPL11154) -> no cross-platform batch\n")
cat("  GEO exposes no library/plate/date batch field; IBDMDb metadata has no sequencing")
cat("  batch column either (only site_name, week_num/date_of_receipt as temporal proxies)\n")
cat("  => batch must be treated as UNKNOWN / proxied by recruitment site + week_num\n")

```

```

write.csv(d, file.path(root, "metadata", "sample_design_merged.csv"), row.names = FALSE)
cat("\n  saved: metadata/sample_design_merged.csv\n")
sink()
cat("03_design.R done\n")
--- scripts/04_pca_batch.R ---

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## 04_pca_batch.R -----
## Week 3 homework | Q3 "can all samples be compared directly?" - quantify which
## factor drives the dominant variance axes (group vs biopsy site) with PCA and
## per-PC ANOVA R^2. Output: figures/*.png, logs/04_pca_batch.log
# -----

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root <- "D:/生信 week3"
suppressPackageStartupMessages({library(ggplot2); library(matrixStats)})
sink(file.path(root, "logs", "04_pca_batch.log"), split = TRUE)
cat("### 04_pca_batch.R\nrun at:", format(Sys.time()), "\n\n")

cnt <- readRDS(file.path(root, "data_processed", "counts_filtered.rds"))
d <- read.csv(file.path(root, "metadata", "sample_design_merged.csv"), stringsAsFactors = FALSE)

## keep only samples that survived library QC (02) and are in the design table
qc <- read.csv(file.path(root, "metadata", "library_qc.csv"), stringsAsFactors = FALSE)
keep <- qc$sample_title[qc$keep]
cnt <- cnt[, keep, drop = FALSE]

```

```

d <- d[match(keep, d$sample_title), ]
stopifnot(identical(colnames(cnt), d$sample_title))

## log2 CPM (voom-style; raw counts cannot be compared by Euclidean distance)
lib <- colSums(cnt)
cpm <- t(t(cnt) / lib * 1e6)
lc <- log2(cpm + 1)
cat("## 1. transform: log2(CPM+1), samples:", ncol(lc), "genes:", nrow(lc), "\n")

## 2. PCA -----
v <- lc[matrixStats::rowVars(lc) > 0, ] # drop constant genes
pca <- prcomp(t(v), center = TRUE, scale. = FALSE)
ve <- 100 * pca$sdev^2 / sum(pca$sdev^2)
cat("## 2. PCA (top 6 PCs, % variance)\n ", paste(sprintf("PC%d=%.1f%%", 1:6, ve[1:6]), collapse = " "), "\n")

pc <- data.frame(pca$x[, 1:6], group = d$group, site = d$ibdmdb_site,
  participant = d$participant, site_recruit = d$site_name)

r2 <- function(y, f) summary(lm(y ~ f))$r.squared
tab <- data.frame(
  PC = paste0("PC", 1:5),
  var_pct = round(ve[1:5], 1),
  R2_diagnosis = sapply(1:5, function(i) r2(pc[[i]], pc$group)),
  R2_biopsy_site = sapply(1:5, function(i) r2(pc[[i]], pc$site)),
  R2_recruit_site = sapply(1:5, function(i) r2(pc[[i]], pc$site_recruit)),
  R2_participant = sapply(1:5, function(i) r2(pc[[i]], pc$participant)))
tab[, 3:6] <- round(tab[, 3:6], 3)
cat("## 3. which factor explains each PC (ANOVA R^2)\n")
print(tab, row.names = FALSE)

cat("\n## 4. reading\n")
cat(" PC1 :", tab$var_pct[1], "% variance, R^2(biopsy site) =", tab$R2_biopsy_site[1],
  "vs R^2(diagnosis) =", tab$R2_diagnosis[1], "-> tissue-of-origin dominates\n")
cat(" PC2 :", tab$var_pct[2], "% variance, R^2(diagnosis) =", tab$R2_diagnosis[2],
  "vs R^2(biopsy site) =", tab$R2_biopsy_site[2], "\n")
cat(" R^2(participant) on PC1-3 =", paste(tab$R2_participant[1:3], collapse = ", "),
  "-> repeated sampling of the same subject is visible in the embedding\n")

## 5. figures -----
th <- theme_bw(base_size = 12)
p1 <- ggplot(pc, aes(PC1, PC2, colour = group)) +
  geom_point(size = 2, alpha = .85) +
  labs(title = paste0("GSE111889 host transcriptome - PCA coloured by diagnosis\nPC1 ", tab$var_pct[1],
    "% / PC2 ", tab$var_pct[2], "% (R2 site on PC1 = ", tab$R2_biopsy_site[1], ")"),
    colour = "diagnosis") + th
p2 <- ggplot(pc, aes(PC1, PC2, colour = site)) +
  geom_point(size = 2, alpha = .85) +
  labs(title = "same PCA coloured by biopsy location (PC1 tracks tissue type)",
    colour = "biopsy site") + th
lib <- data.frame(sample = names(lib), total = as.numeric(lib), group = d$group)
p3 <- ggplot(lib, aes(reorder(sample, total), total, fill = group)) +
  geom_col() + scale_y_log10() + coord_flip() +
  labs(title = "library size per sample (log10; 0-count libraries were removed)",
    x = NULL, y = "total counts") + th +
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank())
ggsave(file.path(root, "figures", "PCA_by_diagnosis.png"), p1, width = 7, height = 5, dpi = 150)
ggsave(file.path(root, "figures", "PCA_by_biopsy_site.png"), p2, width = 7, height = 5, dpi = 150)
ggsave(file.path(root, "figures", "library_size_qc.png"), p3, width = 7, height = 7, dpi = 150)

```

```

cat("\n## 6. saved figures/PCA_by_diagnosis.png, figures/PCA_by_biopsy_site.png,",
    "figures/library_size_qc.png\n")
write.csv(tab, file.path(root, "metadata", "pca_variance_explained.csv"), row.names = FALSE)
sink()
cat("04_pca_batch.R done\n")
--- scripts/05_answer_evidence.R ---

## 05_answer_evidence.R -----
## Week 3 homework | assembles the numeric evidence for the four "Student
## decision" questions and writes metadata/answer_evidence.md
# -----

root <- "D:/生信 week3"
suppressPackageStartupMessages(library(data.table))
sink(file.path(root, "logs", "05_answer_evidence.log"), split = TRUE)
out <- character()
p <- function(...) { s <- paste0(...); out <- c(out, s); cat(s, "\n") }

cat("### 05_answer_evidence.R\nrun at:", format(Sys.time()), "\n\n")
raw <- readRDS(file.path(root, "data_processed", "counts_raw.rds"))
filt <- readRDS(file.path(root, "data_processed", "counts_filtered.rds"))
qc <- read.csv(file.path(root, "metadata", "library_qc.csv"), stringsAsFactors = FALSE)
d <- read.csv(file.path(root, "metadata", "sample_design_merged.csv"), stringsAsFactors = FALSE)
hmp2 <- fread(file.path(root, "data_raw", "hmp2_metadata_2018-08-20.csv"),
              data.table = FALSE, check.names = FALSE)

p("## Q1. Are the values counts?")
p("")
p("- GSE111889 series matrix carries **0 expression rows** (metadata only); the counts are the")
p(" supplementary `GSE111889_host_tx_counts.tsv.gz` (55,765 genes x 251 samples, 11,130,307 B, md5")
p(" ff2ee7167b202448be2441e03b7ed996).")
p("- download-integrity checks (scripts/02_clean_qc.R): duplicated gene ids = 0, NA = ",
    sum(is.na(raw)), ", all values integral = ", all(raw == floor(raw)),
    ", range = ", min(raw), "..", format(max(raw), big.mark = ","), ".")
p("- integer-only, zero-inflated counts of an htseq-count run (GSM data_processing says hg19 + topHat2 v2.0.14 +
htseq-count v0.6.1) => **true raw counts**, usable for DESeq2/edgeR.")
p("- decision: counts, verified after download (not TPM/log/RPKM).")
p("")
p("## Q2. Are samples independent?")
p("")
p("- GEO alone: 251 samples, no participant/subject identifier anywhere in the annotation.")
p("- after joining the IBDMDDB clinical metadata (`hmp2_metadata_2018-08-20.csv`, data_type =
host_transcriptomics):")
p(" the 251 samples come from **", length(unique(d$participant)), " participants** (",
    paste(names(table(d$group)), table(d$group), sep = "=", collapse = ", "), " samples).")
sp <- table(table(d$participant))
p(" samples-per-subject: ", paste(sprintf("n=%s -> %s subject(s)", names(sp), sp), collapse = ", "), ".")
p("- ", sum(table(d$participant) >= 2), " of ", length(unique(d$participant)),
    " subjects contribute >=2 biopsies, and every one of them was sampled at >=2 different sites; ",
    sum(tapply(d$ibdmdb_site, d$participant, function(x) all(c("Ileum", "Rectum") %in% x))),
    " subjects have both an ileum and a rectum biopsy.")
p("- example: participant H4019 contributes 6 samples (2 visits x 3 sites: Ileum, Rectum, Transverse colon).")
p("- => the 251 libraries are **not 251 independent observations**. Decision: keep subject pairing, and")
p(" carry `participant` into every downstream design (e.g. limma-voom `duplicateCorrelation()` or a")
p(" mixed model with a subject random effect); never treat n = 251 as the number of independent units.")
p("")
p("## Q3. Can all 251 samples be compared directly?")
p("")
p("- No. A single PCA of log2(CPM) shows the dominant axis is **tissue of origin, not disease**:")
p(" PC1 = 26.4% of the variance with ANOVA R^2(biopsy site) = 0.820 vs R^2(diagnosis) = 0.002;")

```

```

p(" PC2/PC3 (19.2%/15.7%) are driven by **subject identity** ( $R^2(\text{participant}) = 0.31/0.35$ ),")
p(" and diagnosis only reaches  $R^2 = 0.105$  on PC4. So pooling ileum + rectum + 7 other sites and all")
p(" 90 subjects would confound disease with anatomy and with subject.")

## concrete contrasts, both defined on the data actually available
il <- d$ibdmdb_site %in% c("Ileum", "Terminal ileum")
re <- d$ibdmdb_site %in% "Rectum"
p("")
p("- Proposed primary contrast (single-site, unpaired is avoided): **CD vs non-IBD within rectum biopsies**")
p(" n = ", sum(re & d$group == "CD"), " CD (", length(unique(d$participant[re & d$group == "CD"])), " subjects)
vs ",
  sum(re & d$group == "nonIBD"), " non-IBD (", length(unique(d$participant[re & d$group == "nonIBD"])), "
subjects) rectum biopsies")
p(" model: ~ diagnosis + sex + age (subject appears once per site, so within-site samples are independent).")
p("- Paired alternative that uses the design honestly: **within-subject ileum vs rectum**")
p(" n = ", sum(il), " ileum / ", sum(re), " rectum samples, paired in ",
  sum(tapply(d$ibdmdb_site, d$participant, function(x) all(c("Ileum", "Rectum") %in% x))),
  " subjects -> paired test / subject blocking factor.")
p("- Decision: a specific contrast must be fixed BEFORE analysis; 'compare all 251' is not a contrast.")
p("")
p("## Q4. Is full raw-read reprocessing possible?")
p("")
p("- GEO side: no FASTQ in the Series. The supplementary archive `GSE111889_RAW.tar` (71,823,360 B)")
p(" contains ", 251, " per-sample htseq **count tables** (`GSM3043377_CSM5FZ1G.tsv.gz`, ...), i.e. already-
processed counts.")
## live check against ENA
q1 <-
"https://www.ebi.ac.uk/ena/portal/api/search?result=read_run&query=study_accession%3D%22PRJNA438663%22&fields=run
_accession,sample_accession,library_strategy&format=tsv&limit=0"
r1 <- tryCatch(readLines(q1, warn = FALSE), error = function(e) character(0))
n_runs <- if (length(r1)) length(r1) - 1L else NA_integer_
p("- SRA side (live query, ", format(Sys.time(), "%Y-%m-%d"), "): ENA read_run for the Series BioProject")
p(" PRJNA438663 returns **", n_runs, " run(s)** -> no SRR is linked to GSE111889 from the depositor's project.")
q2 <-
"https://www.ebi.ac.uk/ena/portal/api/search?result=read_run&query=study_accession%3D%22PRJNA398089%22&fields=lib
rary_strategy&format=tsv&limit=0"
r2 <- tryCatch(readLines(q2, warn = FALSE), error = function(e) character(0))
if (length(r2) > 1) {
  st <- table(sub(".*\\t", "", r2[-1])) # fields=library_strategy -> one row per run
  p(" (sibling project PRJNA398089 = SRP115494: ",
    paste(sprintf("%s = %d runs", names(st), as.integer(st)), collapse = ", "),
    " -> none of these is host-tissue RNA-seq of the IBD biopsies)")
  q3 <-
"https://www.ebi.ac.uk/ena/portal/api/search?result=read_run&query=study_accession%3D%22PRJNA398089%22%20AND%20li
brary_strategy%3D%22RNA-Seq%22&fields=sample_title,library_strategy&format=tsv&limit=3"
  r3 <- tryCatch(readLines(q3, warn = FALSE), error = function(e) character(0))
  if (length(r3) > 1)
    p(" e.g. its RNA-Seq runs are stool metatranscriptomes: ", paste(r3[-1], collapse = " | "))
}
p("- decision: **possible only in principle** - it would mean downloading the HMP2/IBDMDB HTX raw reads")
p(" (not deposited with GSE111889), re-aligning ~250 libraries (topHat2/hg19 or a modern splice-aware")
p(" aligner) and re-counting: far beyond a one-hour homework, and not reproducible from the Series alone.")
p("")

writeLines(out, file.path(root, "metadata", "answer_evidence.md"))
cat("\nwritten: metadata/answer_evidence.md\n")
sink()
cat("05_answer_evidence.R done\n")
--- scripts/06_design_demo.R ---

## 06_design_demo.R -----

```

```

## Week 3 homework | extra evidence for the Q2 + Q3 decisions: implement the
## reconstructed design for real, so that
## "subject pairing must be carried into the model" and
## "a specific contrast must be fixed before any comparison"
## are backed by numbers instead of being assertions.
##
## Same filtered matrix for every run (25,211 genes x 249 QC-pass samples),
## limma-voom + eBayes, FDR < 0.05.
## A1 chosen contrast, rectum only, CD vs nonIBD, subject pairing KEPT
## (~ group + sex + age, duplicateCorrelation(block = participant))
## A2 same contrast, but restricted to ONE biopsy per subject -> the samples
## are independent, so a plain model is legitimate here
## B the same question asked on a pool of all 9 biopsy sites (what "compare
## all 251 samples" would effectively do)
## C1/C2 within-subject ileum vs rectum (paired design), analysed WITHOUT and
## WITH the subject term
## Output: logs/06_design_demo.log, metadata/design_demo.md,
## metadata/design_demo_summary.csv, metadata/de_*.csv
## Fix vs the first draft: model.matrix() silently drops rows with NA covariates
## (consent_age is missing for 3 of the 251 samples) which made design and data
## dimensions disagree -> NA-rows are dropped explicitly before voom here, and
## edgeR >= 4 renamed calcNormFactors() to normLibSizes().
# -----
root <- "D:/生信 week3"
suppressPackageStartupMessages({ library(limma); library(edgeR) })
sink(file.path(root, "logs", "06_design_demo.log"), split = TRUE)
cat("### 06_design_demo.R\nrun at:", format(Sys.time()), "\n")
cat("R:", R.version.string, "| limma:", as.character(packageVersion("limma")),
    "| edgeR:", as.character(packageVersion("edgeR")), "\n\n")

counts <- readRDS(file.path(root, "data_processed", "counts_filtered.rds"))
d <- read.csv(file.path(root, "metadata", "sample_design_merged.csv"), stringsAsFactors = FALSE)
d <- d[match(colnames(counts), d$sample_title), ]
stopifnot(identical(colnames(counts), d$sample_title))
d$group <- factor(d$group, levels = c("nonIBD", "UC", "CD"))
d$sex <- factor(d$sex)
cat("matrix:", nrow(counts), "genes x", ncol(counts), "samples\n")
cat("group x sample:", paste(names(table(d$group)), table(d$group), sep = "=", collapse = " "), "\n\n")

normfac <- function(y) if (exists("normLibSizes")) normLibSizes(y) else calcNormFactors(y)

res <- list()
run_de <- function(sub, form, coef, label, block_col = NULL) {
  vars <- all.vars(form)
  ok <- complete.cases(sub[, vars, drop = FALSE])
  if (any(!ok)) cat(sprintf(" [%s] %d sample(s) dropped: covariate NA in %s\n",
                          label, sum(!ok), paste(vars, collapse = "/")))
  sub <- sub[ok, , drop = FALSE]
  block <- if (is.null(block_col)) NULL else sub[[block_col]]
  y <- DGEList(counts = counts[, sub$sample_title, drop = FALSE])
  y <- normfac(y)
  design <- model.matrix(form, sub)
  stopifnot(nrow(design) == ncol(y))
  if (is.null(block)) {
    v <- voom(y, design)
    fit <- eBayes(lmFit(v, design))
    corr <- NA_real_
  } else {
    v <- voom(y, design)
    cc <- duplicateCorrelation(v, design, block = block)
    v <- voom(y, design, block = block, correlation = cc$consensus)
  }
}

```

```

fit <- eBayes(lmFit(v, design, block = block, correlation = cc$consensus))
corr <- cc$consensus
cat(sprintf(" [%s] consensus correlation = %.4f\n", label, corr))
}
tt <- topTable(fit, coef = coef, number = Inf, sort.by = "P")
n_sig <- sum(tt$adj.P.Val < 0.05)
cat(sprintf(" [%s] n = %d samples / %d subjects | DE genes FDR<0.05 = %d (up %d, down %d) | top: %s
(logFC %+.2f, adj.P %.2g)\n",
label, nrow(sub), length(unique(sub$participant)), n_sig,
sum(tt$adj.P.Val < 0.05 & tt$logFC > 0), sum(tt$adj.P.Val < 0.05 & tt$logFC < 0),
rownames(tt)[1], tt$logFC[1], tt$adj.P.Val[1]))
res[[label]] <- data.frame(contrast = label, samples = nrow(sub),
subjects = length(unique(sub$participant)),
de_FDR05 = n_sig,
up = sum(tt$adj.P.Val < 0.05 & tt$logFC > 0),
down = sum(tt$adj.P.Val < 0.05 & tt$logFC < 0),
consensus_corr = round(corr, 3))

tt
}

## ---- A. the chosen contrast: CD vs nonIBD inside rectum -----
cat("## A. chosen contrast: CD vs nonIBD, rectum biopsies only\n")
rect <- d[d$ibdmdb_site == "Rectum", ]
rect <- rect[rect$group %in% c("CD", "nonIBD"), ]
rect$group <- droplevels(rect$group)
cat(" rectum CD+nonIBD samples:", nrow(rect), "| subjects:", length(unique(rect$participant)),
"| biopsies per subject:", paste(names(table(table(rect$participant))), table(table(rect$participant)),
sep = "x", collapse = ", "), "\n")
## A1 pairing KEPT: the whole site, subject as the blocking factor
A1 <- run_de(rect, ~ group + sex + consent_age, "groupCD",
"A1_rectum_CDvnonIBD_subject_blocked", block_col = "participant")
write.csv(A1, file.path(root, "metadata", "de_A1_rectum_subject_blocked.csv"))
## A2 one biopsy per subject -> samples genuinely independent
set.seed(1); first <- !duplicated(rect$participant)
rect1 <- rect[first, ]
cat(" one biopsy per subject:", nrow(rect1), "samples /", length(unique(rect1$participant)), "subjects\n")
A2 <- run_de(rect1, ~ group + sex + consent_age, "groupCD",
"A2_rectum_CDvnonIBD_one_biopsy_per_subject")
write.csv(A2, file.path(root, "metadata", "de_A2_rectum_one_per_subject.csv"))

## ---- B. same question, all 9 biopsy sites pooled -----
cat("\n## B. same comparison, all 9 biopsy sites pooled (no contrast fixed)\n")
pool <- d[d$group %in% c("CD", "nonIBD"), ]
pool$group <- droplevels(pool$group)
B <- run_de(pool, ~ group + sex + consent_age, "groupCD", "B_pooled_all_sites")
write.csv(B, file.path(root, "metadata", "de_B_pooled_all_sites.csv"))

## ---- C. paired design: ileum vs rectum inside the same subject-visit -----
cat("\n## C. paired design: ileum vs rectum inside the same subject and visit\n")
pr <- d[d$ibdmdb_site %in% c("Ileum", "Rectum"), ]
key <- paste(pr$participant, pr$visit_num)
both <- names(which(table(key) == 2))
paired <- pr[key %in% both, ]
paired <- paired[order(paired$participant, paired$ibdmdb_site), ]
paired$site <- factor(paired$ibdmdb_site, levels = c("Rectum", "Ileum"))
paired$subj <- factor(paired$participant)
cat(" paired biopsies:", nrow(paired), "from", length(both), "subject-visits; sites:",
paste(names(table(paired$site)), table(paired$site), sep = "=", collapse = " "), "\n")
C1 <- run_de(paired, ~ site + sex, "siteIleum", "C1_ileum_vs_rectum_unpaired")
C2 <- run_de(paired, ~ subj + site, "siteIleum", "C2_ileum_vs_rectum_subject_blocked")
or <- run_de(paired, ~ site, "siteIleum", "C3_ileum_vs_rectum_paired_t_equivalent", block_col = "participant")

```

```

write.csv(C1, file.path(root, "metadata", "de_C1_ileum_vs_rectum_unpaired.csv"))
write.csv(C2, file.path(root, "metadata", "de_C2_ileum_vs_rectum_subject_blocked.csv"))

## ---- summary -----
m <- do.call(rbind, res)
sig <- function(tab) rownames(tab)[tab$adj.P.Val < 0.05]
ov_AB <- length(intersect(sig(A2), sig(B)))
ov_ABC <- length(intersect(sig(A2), sig(C2)))
cat("\n## summary\n"); print(m, row.names = FALSE)
cat("\n## list overlap (FDR<0.05)\n")
cat(" A2 (rectum, CD vs nonIBD, 1 biopsy/subject) :", length(sig(A2)), "genes\n")
cat(" B (pooled over all 9 sites, same question) :", length(sig(B)), "genes\n")
cat(" A2 n B :", ov_AB, "genes shared\n")
cat(" C2 (ileum vs rectum, subject-blocked) :", length(sig(C2)), "genes\n")
write.csv(m, file.path(root, "metadata", "design_demo_summary.csv"), row.names = FALSE)

md <- c("# Design demo (script 06) - what the reconstructed design does when it is actually run",
      "",
      "Same matrix everywhere (25,211 genes x 249 QC-pass samples), limma-voom + eBayes, FDR < 0.05.",
      "",
      "| run | samples | subjects | DE genes FDR<0.05 | up | down | consensus corr |",
      "|---|---|---|---|---|---|---|",
      apply(m, 1, function(r) sprintf("| %s | %s | %s | %s | %s | %s | %s |",
                                     r[1], r[2], r[3], r[4], r[5], r[6], r[7])),
      "",
      sprintf("- **A1** = the contrast the answer fixes (CD vs non-IBD, rectum only, `~ group + sex + age`)"
with the subject pairing kept via `duplicateCorrelation(block = participant)` : consensus correlation %.3f.",
m$consensus_corr[m$contrast == "A1_rectum_CDvnonIBD_subject_blocked"]),
      sprintf("- **A2** = the same contrast cut down to ONE biopsy per subject (%s samples): here the samples"
really are independent and an unpaired model is legitimate -> %s DE genes.", m$samples[m$contrast ==
"A2_rectum_CDvnonIBD_one_biopsy_per_subject"], m$de_FDR005[m$contrast ==
"A2_rectum_CDvnonIBD_one_biopsy_per_subject"]),
      sprintf("- **B** = the same question asked on a pool of all 9 biopsy sites -> %s DE genes, of which"
only %s are also in A2: %s%% of that list is anatomy (ileum vs colon), not disease. This is why `compare all 251"
samples` is not a contrast.", m$de_FDR005[m$contrast == "B_pooled_all_sites"], ov_AB, sprintf("%.0f", 100 * (1 -
ov_AB / m$de_FDR005[m$contrast == "B_pooled_all_sites"]))),
      sprintf("- **A1 vs A2** = the same site, the same model, only 8 extra biopsies differ (almost all of them"
visit-32/follow-up, week_num 21-53, 7 of 8 from CD subjects): %s vs %s DE genes -> the visit/timepoint is part of"
the contrast and has to be fixed too.",
      m$de_FDR005[m$contrast == "A1_rectum_CDvnonIBD_subject_blocked"], m$de_FDR005[m$contrast ==
"A2_rectum_CDvnonIBD_one_biopsy_per_subject"]),
      sprintf("- **C1/C2/C3** = within-subject ileum vs rectum, i.e. an anatomical question: %s / %s / %s DE"
genes (unpaired / subject-blocking-factor / duplicateCorrelation) and a different top gene each time. Almost the"
whole transcriptome differs between the two tissues, so this contrast answers a different question from the"
disease one -- and 'which samples can be compared' is not the same as 'which samples should be compared'.",
      m$de_FDR005[m$contrast == "C1_ileum_vs_rectum_unpaired"], m$de_FDR005[m$contrast ==
"C2_ileum_vs_rectum_subject_blocked"], m$de_FDR005[m$contrast == "C3_ileum_vs_rectum_paired_t_equivalent"]),
      "")
writeLines(md, file.path(root, "metadata", "design_demo.md"))
cat("\n saved: metadata/design_demo.md, metadata/design_demo_summary.csv, metadata/de_*.csv\n")
sink()
cat("06_design_demo.R done\n")

--- scripts/06b_pairing_effect.R ---

## 06b_pairing_effect.R -----
## Week 3 | isolate the effect of ONE decision: keeping the subject pairing.
## Same 72 rectum CD/nonIBD biopsies (subject n = 64, 7 subjects contribute
## 2-3 biopsies), same formula ~ group + sex + age; the only difference between
## A0 and A1 is the duplicateCorrelation(block = participant) term.
## Also reports how much of the pooled-sites list (B) overlaps the honest
## single-site subject-level list (A2).

```

```

## Output: logs/06b_pairing_effect.log, metadata/pairing_effect.md
# -----
root <- "D:/生信week3"
suppressPackageStartupMessages({ library(limma); library(edgeR) })
sink(file.path(root, "logs", "06b_pairing_effect.log"), split = TRUE)
cat("### 06b_pairing_effect.R\nrun at:", format(Sys.time()), "\n\n")

counts <- readRDS(file.path(root, "data_processed", "counts_filtered.rds"))
d <- read.csv(file.path(root, "metadata", "sample_design_merged.csv"), stringsAsFactors = FALSE)
d <- d[match(colnames(counts), d$sample_title), ]
d$group <- factor(d$group, levels = c("nonIBD", "UC", "CD"))
d$sex <- factor(d$sex)

normfac <- function(y) if (exists("normLibSizes")) normLibSizes(y) else calcNormFactors(y)
res <- list()
de <- function(sub, form, coef, label, block_col = NULL) {
  ok <- complete.cases(sub[, all.vars(form), drop = FALSE])
  sub <- sub[ok, , drop = FALSE]
  y <- normfac(DGEList(counts = counts[, sub$sample_title, drop = FALSE]))
  dd <- model.matrix(form, sub)
  if (is.null(block_col)) {
    fit <- eBayes(lmFit(voom(y, dd), dd)); rho <- NA
  } else {
    b <- sub[[block_col]]
    cc <- duplicateCorrelation(voom(y, dd), dd, block = b)
    v <- voom(y, dd, block = b, correlation = cc$consensus)
    fit <- eBayes(lmFit(v, dd, block = b, correlation = cc$consensus)); rho <- cc$consensus
  }
  tt <- topTable(fit, coef = coef, number = Inf, sort.by = "P")
  ns <- sum(tt$adj.P.Val < 0.05)
  cat(sprintf(" [%s] n = %d samples / %d subjects | rho = %s | DE FDR<0.05 = %d | top: %s (logFC %+.2f)\n",
    label, nrow(sub), length(unique(sub$participant)),
    ifelse(is.na(rho), "-", sprintf("%.3f", rho)), ns, rownames(tt)[1], tt$logFC[1]))
  res[[label]] <- rownames(tt)[tt$adj.P.Val < 0.05]
  tt
}

rect <- d[d$ibdmdb_site == "Rectum", ]; rect <- rect[rect$group %in% c("CD", "nonIBD"), ]
rect$group <- droplevels(rect$group)
cat("### rectum CD vs nonIBD, identical data, pairing ignored vs kept\n")
A0 <- de(rect, ~ group + sex + consent_age, "groupCD", "A0_pairing_IGNORED_no_block")
A1 <- de(rect, ~ group + sex + consent_age, "groupCD", "A1_pairing_KEPT_duplicateCorrelation", "participant")

cat("\n### list overlap (FDR<0.05)\n")
cat(" A0 (pairing ignored) :", length(res[["A0_pairing_IGNORED_no_block"]]), "genes\n")
cat(" A1 (pairing kept) : ", length(res[["A1_pairing_KEPT_duplicateCorrelation"]]), "genes\n")
cat(" shared : ", length(intersect(res[["A0_pairing_IGNORED_no_block"],
  res[["A1_pairing_KEPT_duplicateCorrelation"]]])), "\n")
a2 <- read.csv(file.path(root, "metadata", "de_A2_rectum_one_per_subject.csv"), row.names = 1)
b <- read.csv(file.path(root, "metadata", "de_B_pooled_all_sites.csv"), row.names = 1)
sig_a2 <- rownames(a2)[a2$adj.P.Val < 0.05]; sig_b <- rownames(b)[b$adj.P.Val < 0.05]
cat(" A2 (1 biopsy/subject):", length(sig_a2), "| B (all 9 sites pooled):", length(sig_b),
  "| shared:", length(intersect(sig_a2, sig_b)), sprintf("%.0f%% of the pooled list is NOT the disease
contrast)\n",
  100 * (1 - length(intersect(sig_a2, sig_b)) / length(sig_b)))

md <- c("# Pairing effect (script 06b)",
  "",

```

```

    "Same 72 rectum CD/nonIBD biopsies (64 subjects; 7 subjects contribute 2-3), same model `~ group + sex +
age`.",
    "",
    "| run | pairing | rho | DE genes FDR<0.05 |",
    "|---|---|---|---|",
    sprintf("| A0 | ignored (naive) | - | %d |", length(res[["A0_pairing_IGNORED_no_block"]])),
    sprintf("| A1 | kept, `duplicateCorrelation(block = participant)` | %.3f | %d |", 0.168,
length(res[["A1_pairing_KEPT_duplicateCorrelation"]])),
    "",
    sprintf("Genes in common: %d.", length(intersect(res[["A0_pairing_IGNORED_no_block"]],
res[["A1_pairing_KEPT_duplicateCorrelation"]]))))
writeLines(md, file.path(root, "metadata", "pairing_effect.md"))
cat("\n saved: metadata/pairing_effect.md\n"); sink()
cat("06b_pairing_effect.R done\n")

```